

ISTRANA, Italy

WMO: 160980

Lat: 45.685N

Lon: 12.083E

Elev: 42

StdP: 100.82

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.9	-3.2	-12.3	1.3	1.7	-9.2	1.7	1.5	8.5	5.2	7.0	5.1	1.3	40	0.409

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	34.1	23.6	32.7	22.9	31.2	22.1	25.4	30.5	24.4	29.6	23.5	28.8	2.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.0	28.0	22.9	17.8	27.1	21.9	16.6	26.3	78.1	30.4	73.8	29.2	70.2	28.7	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.1	5.2	-7.2	36.0	2.3	1.5	-8.8	37.1	-10.2	38.0	-11.5	38.8	-13.1	39.9		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-7.8	27.0	2.3	1.5	-9.5	28.0	-10.8	28.9	-12.1	29.7	-13.8	30.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.4	3.7	5.2	9.5	13.9	18.7	22.8	25.0	24.6	19.8	14.7	9.2	4.6
	DBStd	8.13	2.76	2.92	3.10	3.17	3.08	3.11	2.49	2.95	2.97	3.24	3.48	3.18
	HDD10.0	607	194	134	46	5	0	0	0	0	0	4	54	169
	HDD18.3	2104	452	367	274	136	33	3	0	1	19	118	274	427
	CDD10.0	2199	1	1	29	122	270	385	464	453	293	150	31	1
	CDD18.3	654	0	0	0	3	44	139	206	195	62	6	0	0
	CDH23.3	5962	0	0	1	34	370	1266	2036	1859	383	13	0	0
	CDH26.7	2304	0	0	0	3	89	471	858	798	84	1	0	0
Wind	WSAvg	1.9	1.6	2.0	2.2	2.4	2.2	2.1	2.0	1.9	1.8	1.6	1.7	1.6
Precipitation	PrecAvg	958	50	51	61	87	100	92	78	75	95	102	101	67
	PrecMax	1675	259	240	270	147	223	184	209	155	181	194	208	163
	PrecMin	660	0	0	1	3	28	28	41	17	27	7	22	5
	PrecStd	243	53	61	59	38	54	39	36	37	47	54	49	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	15.9	21.6	26.2	31.4	34.9	35.6	36.5	30.8	25.0	18.4	13.4
		MCWB	9.6	9.2	12.5	16.7	20.2	23.4	23.5	23.4	20.7	18.2	14.9	10.8
	2%	DB	10.9	13.5	19.1	24.1	29.1	32.8	33.8	34.6	29.0	23.0	16.7	12.0
		MCWB	8.0	8.1	11.5	15.1	19.3	23.0	24.0	23.3	20.3	17.6	13.9	9.6
	5%	DB	9.7	11.9	17.1	22.2	27.1	31.1	32.2	32.8	27.2	21.3	15.4	10.6
		MCWB	7.2	7.5	10.9	14.5	18.3	22.0	22.9	22.9	19.6	16.8	13.1	8.3
	10%	DB	8.2	10.2	15.2	20.2	25.2	29.7	31.1	31.0	25.6	19.9	14.2	9.2
		MCWB	6.2	7.1	10.0	13.7	17.5	21.3	22.2	22.3	18.8	16.0	12.1	7.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	10.7	14.4	18.0	22.7	26.2	26.9	27.1	23.2	20.3	16.2	12.0
		MCDB	12.1	13.7	18.6	23.4	27.5	31.4	31.9	31.9	27.4	22.3	17.7	12.8
	2%	WB	9.0	9.5	12.9	16.5	20.9	24.4	25.5	25.4	22.0	19.1	14.8	10.2
		MCDB	10.0	11.8	16.5	21.9	26.4	29.9	30.7	30.1	26.2	21.4	16.0	11.2
	5%	WB	7.7	8.6	12.0	15.4	19.6	23.2	24.4	24.4	21.1	18.0	13.7	8.8
		MCDB	8.8	10.7	15.5	20.5	24.9	28.7	29.5	29.4	24.8	20.2	15.0	9.8
	10%	WB	6.5	7.7	11.1	14.5	18.5	22.2	23.5	23.4	20.0	16.9	12.5	7.8
		MCDB	7.7	9.6	14.2	19.0	23.5	27.6	28.7	28.7	24.0	19.0	13.9	8.9
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.8	9.9	10.2	10.4	10.5	10.8	10.7	9.9	8.3	7.1	7.5
		MCDBR	9.3	11.6	13.0	13.1	13.0	12.1	11.9	12.4	11.5	9.7	7.7	8.4
		MCWBR	6.5	7.0	6.9	6.0	5.3	4.7	4.5	4.4	4.7	5.1	5.1	5.9
	5% WB	MCDBR	6.7	8.7	10.5	11.3	11.1	11.1	11.0	11.2	9.8	7.7	5.7	6.2
		MCWBR	5.4	6.0	6.3	5.8	5.4	5.3	5.1	5.0	5.0	5.0	4.4	5.0
Clear-Sky Solar Irradiance	taub	0.344	0.372	0.420	0.464	0.470	0.492	0.475	0.465	0.447	0.428	0.378	0.340	
	taud	2.388	2.294	2.168	2.075	2.112	2.101	2.157	2.199	2.211	2.219	2.312	2.417	
	Ebn at Noon	746	791	798	795	805	787	796	787	763	712	689	710	
	Edh at Noon	75	99	129	154	153	156	145	134	122	106	81	67	
All-Sky Solar Radiation	RadAvg	1.38	2.24	3.52	4.58	5.58	6.34	6.46	5.53	4.05	2.51	1.46	1.20	
	RadStd	0.21	0.36	0.46	0.52	0.55	0.36	0.35	0.42	0.32	0.31	0.21	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (53 neighbors)	+0.68	+0.96	+1.89	N/A	N/A	N/A	-93	-172	+117	N/A
		+0.41	N/A	N/A	+0.60	+0.37	N/A	N/A	-116	+78	+27

Nomenclature: See separate page